

Question: Did Guyana’s oil discovery increase GDP growth relative to CARICOM?

Countries: Barbados, Jamaica, Trinidad & Tobago, Suriname, Belize, and Grenada.
Model:

$$GDP_growth = \beta_0 + \beta_1 did + \text{Year FE} + \text{Country FE} + \epsilon$$

Did = treatment x post

Country FE: controls for differences within countries such as population size, industrial base, political stability, geography, etc.
Removes baseline differences within countries, such as Jamaica growing differently than Trinidad & Tobago.

Year FE: Controls for differences over time.

Data

Panel Data Source & Initial Observation Preview							
Status Message: Panel dataset created successfully!							
File Destination: C:/Users/Ravin/Downloads/Guyana_CARICOM_GDP_long.csv							
Index	Country Name	Country Code	Year	GDP_growth (%)	treatment	post	did
0	Guyana	GUY	2000	-1.363804	1	0	0
1	Barbados	BRB	2000	4.447189	0	0	0
2	Jamaica	JAM	2000	0.878704	0	0	0
3	Trinidad and To- bago	TTO	2000	6.901360	0	0	0
4	Suriname	SUR	2000	2.099999	0	0	0

2. OLS Model Performance & Goodness-of-Fit Summary

Model Parameters		Fit Statistics & Information Criteria	
Dep. Variable:	GDP_growth	R-squared:	0.533
Model:	OLS	Adj. R-squared:	0.433
Method:	Least Squares	F-statistic:	5.313
No. Observations:	182	Prob (F-statistic):	9.35×10^{-13}
Df Residuals:	149	Log-Likelihood:	-575.14
Df Model:	32	AIC:	1216.
Covariance Type:	nonrobust	BIC:	1322.

3. OLS Estimation Results & Coefficients

Variable / Factor	Coefficient	Std. Error	t-statistic	P > t	[95% Conf. Interval]
Intercept	3.1112	2.648	1.175	0.242	[-2.122, 8.344]
Year Fixed Effects (Baseline Year: 2000)					
C(Year)[T.2001]	-2.4993	3.370	-0.742	0.459	[-9.158, 4.159]
C(Year)[T.2002]	-0.9431	3.370	-0.280	0.780	[-7.601, 5.715]

Variable / Factor	Coefficient	Std. Error	t-statistic	$P > t $	[95% Conf. Interval]
C(Year)[T.2003]	2.1411	3.370	0.635	0.526	[-4.517, 8.800]
C(Year)[T.2004]	-0.7443	3.370	-0.221	0.825	[-7.403, 5.914]
C(Year)[T.2005]	-0.1413	3.370	-0.042	0.967	[-6.800, 6.517]
C(Year)[T.2006]	0.5036	3.370	0.149	0.881	[-6.155, 7.162]
C(Year)[T.2007]	-0.0087	3.370	-0.003	0.998	[-6.667, 6.650]
C(Year)[T.2008]	-3.0943	3.370	-0.918	0.360	[-9.753, 3.564]
C(Year)[T.2009]	-6.3259	3.370	-1.877	0.062	[-12.984, 0.333]
C(Year)[T.2010]	-3.7276	3.370	-1.106	0.270	[-10.386, 2.931]
C(Year)[T.2011]	-2.4357	3.370	-0.723	0.471	[-9.094, 4.223]
C(Year)[T.2012]	-2.2996	3.370	-0.682	0.496	[-8.958, 4.359]
C(Year)[T.2013]	-1.8581	3.370	-0.551	0.582	[-8.516, 4.800]
C(Year)[T.2014]	-1.6782	3.370	-0.498	0.619	[-8.337, 4.980]
C(Year)[T.2015]	-6.4545	3.392	-1.903	0.059	[-13.156, 0.248]
C(Year)[T.2016]	-7.4495	3.392	-2.196	0.030	[-14.151, -0.748]
C(Year)[T.2017]	-6.4890	3.392	-1.913	0.058	[-13.191, 0.213]
C(Year)[T.2018]	-5.0732	3.392	-1.496	0.137	[-11.775, 1.629]
C(Year)[T.2019]	-5.3474	3.392	-1.577	0.117	[-12.049, 1.355]
C(Year)[T.2020]	-12.2397	3.392	-3.609	0.000	[-18.942, -5.538]
C(Year)[T.2021]	0.2738	3.392	0.081	0.936	[-6.428, 6.976]
C(Year)[T.2022]	7.7348	3.392	2.281	0.024	[1.033, 14.437]
C(Year)[T.2023]	-0.5912	3.392	-0.174	0.862	[-7.293, 6.111]
C(Year)[T.2024]	0.7689	3.392	0.227	0.821	[-5.933, 7.471]
C(Year)[T.2025]	-3.0174	3.392	-0.890	0.375	[-9.719, 3.685]
Country Fixed Effects (Baseline Country: Barbados)					
C(Country_Name)[T.Belize]	2.4934	1.748	1.426	0.156	[-0.962, 5.948]
C(Country_Name)[T.Grenada]	1.6540	1.748	0.946	0.346	[-1.801, 5.109]
C(Country_Name)[T.Guyana]	1.0074	2.089	0.482	0.630	[-3.121, 5.136]
C(Country_Name)[T.Jamaica]	0.2571	1.748	0.147	0.883	[-3.198, 3.712]
C(Country_Name)[T.Suriname]	1.2595	1.748	0.720	0.472	[-2.195, 4.714]
C(Country_Name)[T.Trinidad and Tobago]	1.6424	1.748	0.939	0.349	[-1.812, 5.097]
Treatment Estimator (Difference-in-Differences Interaction)					
did	21.3116	2.703	7.885	0.000	[15.971, 26.653]

4. Model Residual Diagnostics & Policy Takeaway

Diagnostic Test	Value	Diagnostic Test	Value
Omnibus Test:	71.818	Durbin-Watson Stat:	2.189
Prob(Omnibus):	0.000	Jarque-Bera (JB) Stat:	590.958
Skewness:	1.210	Prob(JB):	4.73×10^{-129}
Kurtosis:	11.489	Condition No.:	28.7

The treatment variable was created to identify which country received the shock

Treatment = 1 for Guyana

Treatment = 0 for all other CARICOM countries

This is needed so that the model can distinguish Guyana from the control Group

Post variable

Needed to confirm when the shock happened

Post = 1 for years $\geq$ 2015

Post = 0 for years $<$ 2015

This allows the model to know the oil discovery happened in 2015

Did = treatment x post

- Guyana (treatment =1)
- After 2015 (post =1)

Did = 1 for Guyana alone after the oil discovery

And 0 for everyone else

Statistical Significance of Oil: Estimated coefficient $\beta_{did} = 21.3116$ ($t = 7.885$, $p < 0.001$), making it statistically significant. Post-discovery, Guyana saw an average additional **21.31** percentage point increase in real GDP growth relative to CARICOM control countries.

Fit: The model accounts for 53.3% of the variance ($R^2 = 0.533$, adjusted $R^2 = 0.433$), it has a highly significant overall (**F-statistic** $P = 9.35 \times 10^{-13} < 0.05$).

A typical question would be why the model includes other CARICOM countries if their individual coefficients are not interpreted directly. One could have examined Guyana's GDP growth from 2000-2025 in isolation, but this would not have allowed for causal inference. Difference-in-Differences requires a control group to establish a counterfactual of what Guyana's growth would have been in the absence of the oil discovery.

The additional CARICOM countries act as the control group. They absorb regional trends, global shocks, and time-invariant country characteristics. Countries like Trinidad & Tobago also have oil sectors, yet none have witnessed a structural transformation comparable to Guyana's post-2015 boom. Including them allows the model to isolate the true impact of the oil discovery rather than confounding it with broader Caribbean or global economic movements.